

A BIQUADRATIC FIXED-FIELD ENVELOPE WITH EXPLICIT PRIMITIVE GENERATORS IN A $W(E_6)$ -EXTENSION

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ABSTRACT. We construct an explicit biquadratic fixed-field envelope inside a normal $W(E_6)$ -extension arising from the 27 lines on a cubic surface. Starting from a degree-320 Gassmann pair, an exhaustive eleven-bucket computation finds that its transported members are the only collision whose normalizers are conjugate in $W(E_6)$ and have index two over both subgroups. Their common normalizer produces fields of degrees 160, 320, and 640: the top is biquadratic over the degree-160 field, its three quadratic subfields have a common normal closure, and primitive integral line-coordinate generators are given for the base, a third quadratic companion, and the top. The companion requires a cubic formal invariant: its point and unordered-pair orbit partitions agree with those of the normalizer, ruling out an exact stabilizer in total degree at most two. We also prove a rational permutation-character relation and the resulting zeta identity, compute all signatures and signed discriminants, and obtain relative discriminant norms $3^8, 3^{16}, 3^8, 3^{32}$. Complete double-coset tables for both retained 3-adic decomposition-group branches show that all relative ramification is tame above 3, with $e = 2$ and different exponent 1 in every ramified row. The proof combines Galois correspondence, exact stabilizers, noncollision at the completely split prime 692717, and independent finite certificates; neither local branch is selected.

1. INTRODUCTION AND THE INTEGRATED THEOREM

An equality of permutation characters can make two number fields arithmetically equivalent without explaining how they sit inside their common normal closure. In particular, a Gassmann collision does not by itself produce a common intermediate field, a third quadratic companion, or usable primitive elements. Those are embedding-sensitive questions: they depend on the exact subgroups in the exact permutation action, not merely on abstract group isomorphism types or equal character rows. This paper resolves those questions for one normal $W(E_6)$ -extension attached to the 27 lines of a cubic surface.

The construction begins with a fixed degree-320 Gassmann pair K^{H+} and K^{H-} . We transport the second subgroup into a common normalizer and obtain three index-two subgroups H_+, H_0, H_3 of a group N of order 324. Their common intersection J has order 81, and

$$N/J \cong V_4.$$

Reversing inclusions by Galois correspondence gives a field diamond

$$M = K^N \subset F_+, F_0, F_3 \subset L = K^J.$$

What turns this abstract diamond into an explicit construction is a set of three integral line-coordinate carriers. Quadratic carriers generate M and L , while the new companion F_0 is generated by a cubic orbit sum. The need for degree three is formal and exact: H_0 and N have the same orbits on points and unordered pairs, so their invariant spaces coincide in total degree at most two.

Arithmetic equivalence and its relation to induced trivial characters go back to the classical criterion of Perlis [6, Theorem 1, printed p. 351]; Bosma and de Smit give computational context and discuss invariants forced by the equality [1, printed p. 68]. James's complete $W(E_6)$ subgroup calculation supplies the ambient figures of 350 conjugacy classes, 339

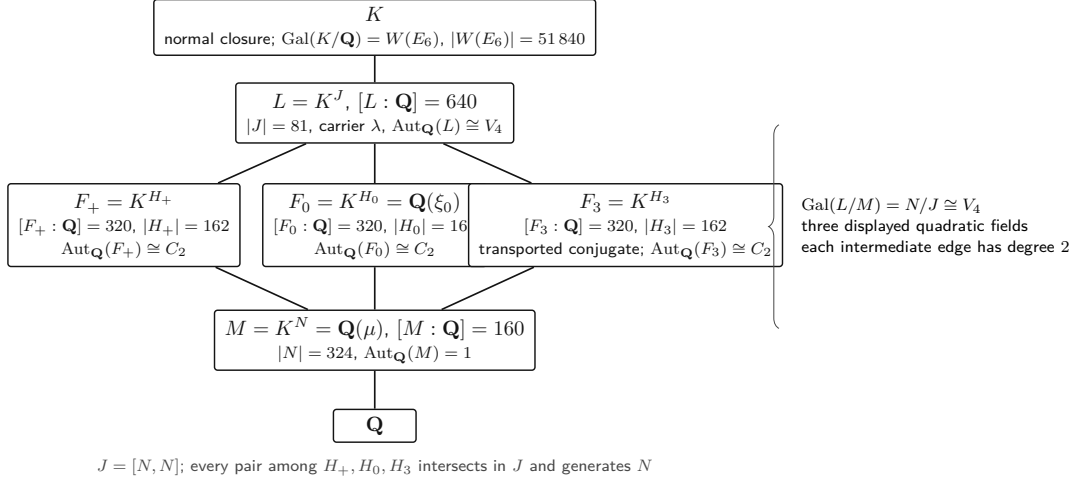


FIGURE 1. The common normalizer produces a biquadratic fixed-field envelope. The three order-162 subgroups H_+, H_0, H_3 contain the common order-81 subgroup $J = [N, N]$, pairwise intersect in J , and pairwise generate the order-324 group N . Thus L/M has Galois group V_4 with exactly the three displayed quadratic intermediate fields; the ambient normal closure is the fixed $W(E_6)$ -extension $K/\mathbf{Q}$. The diagram makes no uniqueness claim beyond this fixed configuration.

permutation-character profiles, and eleven collisions [4, §3.1, printed p. 22]. The present claim is not a new collision criterion and not a globally unique V_4 construction. It is the following exact conjunction for the fixed released action: among those eleven collision buckets, the pair 301/303 alone has normalizers conjugate in $W(E_6)$ with index two over both members, and that embedded configuration supports the explicit field and arithmetic package below. The classical resolvent framework used to prove primitivity is due to Stauduhar and its computational invariant formulation to Fieker and Klüners [7, pp. 981–984] [3, §§1–2 and Remark 2.1]. Explicit cubic-surface realizations of $W(E_6)$ provide the broader geometric context [2].

We now state the result as one theorem. The notation in its statement is defined without software locators in section 2; all finite tables needed to check it appear in the appendices.

Theorem 1.1 (Explicit biquadratic fixed-field envelope). *Let $K/\mathbf{Q}$ be the fixed normal extension of section 2, with $G = \text{Gal}(K/\mathbf{Q}) = W(E_6)$ and $|G| = 51840$. Let H_+ and H_- be the displayed embedded Gassmann subgroups, let $H_3 = xH_-x^{-1}$ under the left label-map convention, and let N, H_0, J be the exact subgroups in section 3. Put*

$$M = K^N, \quad F_i = K^{H_i} \quad (i \in \{+, 0, 3\}), \quad L = K^J.$$

Then the following assertions hold together.

- (i) Lattice. *The orders of N, H_i, J are 324, 162, 81; $J = [N, N]$, $N/J \cong V_4$, every pair of the H_i intersects in J , and every pair generates N . Consequently*

$$[M : \mathbf{Q}] = 160, \quad [F_i : \mathbf{Q}] = 320, \quad [L : \mathbf{Q}] = 640,$$

L/M is biquadratic, and F_+, F_0, F_3 are its three quadratic subfields.

- (ii) Closures and automorphisms. *Every field in the diamond has normal closure K . Moreover*

$$\text{Aut}_{\mathbf{Q}}(M) = 1, \quad \text{Aut}_{\mathbf{Q}}(F_i) \cong C_2, \quad \text{Aut}_{\mathbf{Q}}(L) \cong V_4.$$

The field F_3 is a $\mathbf{Q}$ -conjugate of the original embedded minus field K^{H_-} .

- (iii) Primitive generators. *For the integral scaled line roots $\alpha_1, \dots, \alpha_{27}$, the explicitly defined carriers μ, ξ_0, λ satisfy*

$$M = \mathbf{Q}(\mu), \quad F_0 = \mathbf{Q}(\xi_0), \quad L = \mathbf{Q}(\lambda).$$

Their formal stabilizer orders are 324, 162, and 81, and the corresponding orbit sizes are 160, 320, and 640. All orbit values are distinct after reduction at the unramified completely split prime $p = 692717$.

- (iv) Formal invariant gap. *The H_0 - and N -orbit partitions on points and unordered pairs agree: the point orbit has size 27, and the pair-orbit sizes are*

$$27, 27, 54, 81, 162.$$

Hence no commutative $\mathbf{Q}$ -coefficient formal polynomial of total degree at most two in the 27 labelled variables has stabilizer exactly H_0 . The displayed 27-term cubic carrier ξ_0 does.

- (v) Restricted uniqueness. *Among the eleven frozen collision buckets, 301/303 is the only pair for which the two normalizers are conjugate in G and both normalizer indices over the corresponding subgroups are equal to two. No broader uniqueness of a V_4 configuration is asserted.*
- (vi) Characters and zeta functions. *On all 25 conjugacy classes, the following equality holds in the rational permutation representation ring:*

$$[G/J] + 2[G/N] = [G/H_+] + [G/H_0] + [G/H_3].$$

It follows that

$$\zeta_L(s)\zeta_M(s)^2 = \zeta_{F_+}(s)\zeta_{F_0}(s)\zeta_{F_3}(s) = \zeta_{F_+}(s)^2\zeta_{F_0}(s).$$

The bracket equality is not an isomorphism of finite G -sets.

- (vii) Global and relative arithmetic. *With A, B defined in [equations \(6.8\)](#) and [\(6.10\)](#), the signatures and signed discriminants are those in [table 1](#). The exact finite support is*

$$\{3, 5, 181, 283, 997, 1801, 2346241, \\ 14932047182473291995860108491583652133938007263719\}.$$

The absolute norms of the relative discriminants over M are

$$3^8, \quad 3^{16}, \quad 3^8, \quad 3^{32}$$

for F_+, F_0, F_3, L , respectively.

- (viii) Both retained local towers. *The complete relative tables hold for both retained possibilities $D_3 = \text{ToM } 140$ and $D_3 = \text{ToM } 206$, with relative factor totals (36, 28, 36, 56) and (18, 14, 18, 28). All relative ramification lies above 3; every ramified relative row has $e = 2$ and different exponent $d = 1$, hence is tame. Neither branch is selected.*

The proof has four conceptual moves. First, exact subgroup intersections, normalizers, and cores give the field lattice, normal closures, and automorphisms. Second, stabilizers of three integral formal carriers are upgraded to stabilizers of evaluated elements by split-prime noncollision. Third, the elementary character relation for V_4 is inflated and induced to G . Finally, conductor and double-coset formulas compute global and relative arithmetic independently in both local branches. Finite exhaustions discharge bounded lemmas; they do not replace these mathematical implications.

Our contributions are therefore specific and falsifiable:

- (1) we identify and prove the common-normalizer envelope under the exact eleven-bucket predicate, including its 112/120 near-miss;
- (2) we construct primitive integral generators of the three key levels and prove the quadratic-to-cubic formal invariant gap;
- (3) we derive the rational character and zeta identities and compute the complete signed global and relative discriminants; and

TABLE 1. Arithmetic dashboard for the five fixed fields. Primitive carriers are displayed for the base M , the new companion F_0 , and the top L ; F_3 is the transported conjugate field. The exact products $A = 181 \cdot 997 \cdot 2346241$ and $B = 283 \cdot 1801 \cdot 14932047182473291995860108491583652133938007263719$ partition the six non-3, 5 primes in the inherited eight-prime support. Relative entries are absolute norms of relative discriminant ideals over M .

field	fixed subgroup	degree	carrier	$\text{Aut}_{\mathbf{Q}}$	signature	signed discriminant	$N_{M/\mathbf{Q}}(\mathfrak{d}_{E/M})$
M	N	160	μ	1	(16, 72)	$+3^{308}5^{248}A^{96}B^{80}$	–
F_+	H_+	320	–	C_2	(16, 152)	$+3^{624}5^{496}A^{192}B^{160}$	3^8
F_0	H_0	320	ξ_0	C_2	(0, 160)	$+3^{632}5^{496}A^{192}B^{160}$	3^{16}
F_3	H_3	320	–	C_2	(16, 152)	$+3^{624}5^{496}A^{192}B^{160}$	3^8
L	J	640	λ	V_4	(0, 320)	$+3^{1264}5^{992}A^{384}B^{320}$	3^{32}

$$A = 181 \cdot 997 \cdot 2346241.$$

$$B = 283 \cdot 1801 \cdot 14932047182473291995860108491583652133938007263719.$$

The F_3 row is the transported conjugate field, not a silent identification with the original embedded minus field.

- (4) we prove the same tame relative-ramification conclusion from two separate exhaustive 3-adic towers without selecting between them.

Section 2 fixes the released premise and action convention. Sections 3 and 4 prove the field and generator statements. Sections 5 to 7 prove the uniqueness, character, and arithmetic clauses. Section 8 records the finite evidence and the exact boundary of the claims. Appendices A–D give durable arrays, exhaustive rows, certificate fingerprints, and scope checks.

2. THE RELEASED $W(E_6)$ INPUT AND ACTION CONVENTIONS

This section separates the fixed arithmetic premise from the new envelope. The premise is an exact normal extension $K/\mathbf{Q}$ obtained from the lines of a smooth cubic surface; the new statements begin only after the exact embedded subgroups are placed in its labelled action. This separation matters because an abstract copy of $W(E_6)$ would not identify any of the fixed fields below.

2.1. The cubic surface and labelled line action. Let $Y = V(\Phi) \subset \mathbf{P}_{\mathbf{Q}}^3$, where

$$\begin{aligned} \Phi = & 75081586157u_0^3 - 28576620789u_0^2u_1 - 122000922135u_0^2u_2 - 5364921951u_0^2u_3 \\ & + 164150208636u_0u_1^2 - 415458334296u_0u_1u_2 + 151070718312u_0u_1u_3 \\ & + 1158143874300u_0u_2^2 + 114691988016u_0u_2u_3 + 113572676646u_0u_3^2 \\ & + 6898957820u_1^3 + 1132596902196u_1^2u_2 - 30413540316u_1^2u_3 \\ & - 2054867641020u_1u_2^2 + 151980984216u_1u_2u_3 + 36794420832u_1u_3^2 \\ & + 2646295985484u_2^3 + 560186573940u_2^2u_3 + 706181383584u_2u_3^2 + 1884468968u_3^3. \end{aligned} \tag{2.1}$$

The released line reconstruction supplies a nonmonic eliminant

$$g(d) = L_0d^{27} + b_{26}d^{26} + \cdots + b_0 \in \mathbf{Z}[d]$$

and a fixed ordering $d_1, \dots, d_{27}$ of its roots corresponding to the 27 reconstructed lines. We reserve L for the degree-640 field and use $L_0 > 0$ for this leading coefficient. Set

$$\alpha_i = L_0d_i \quad (1 \leq i \leq 27). \tag{2.2}$$

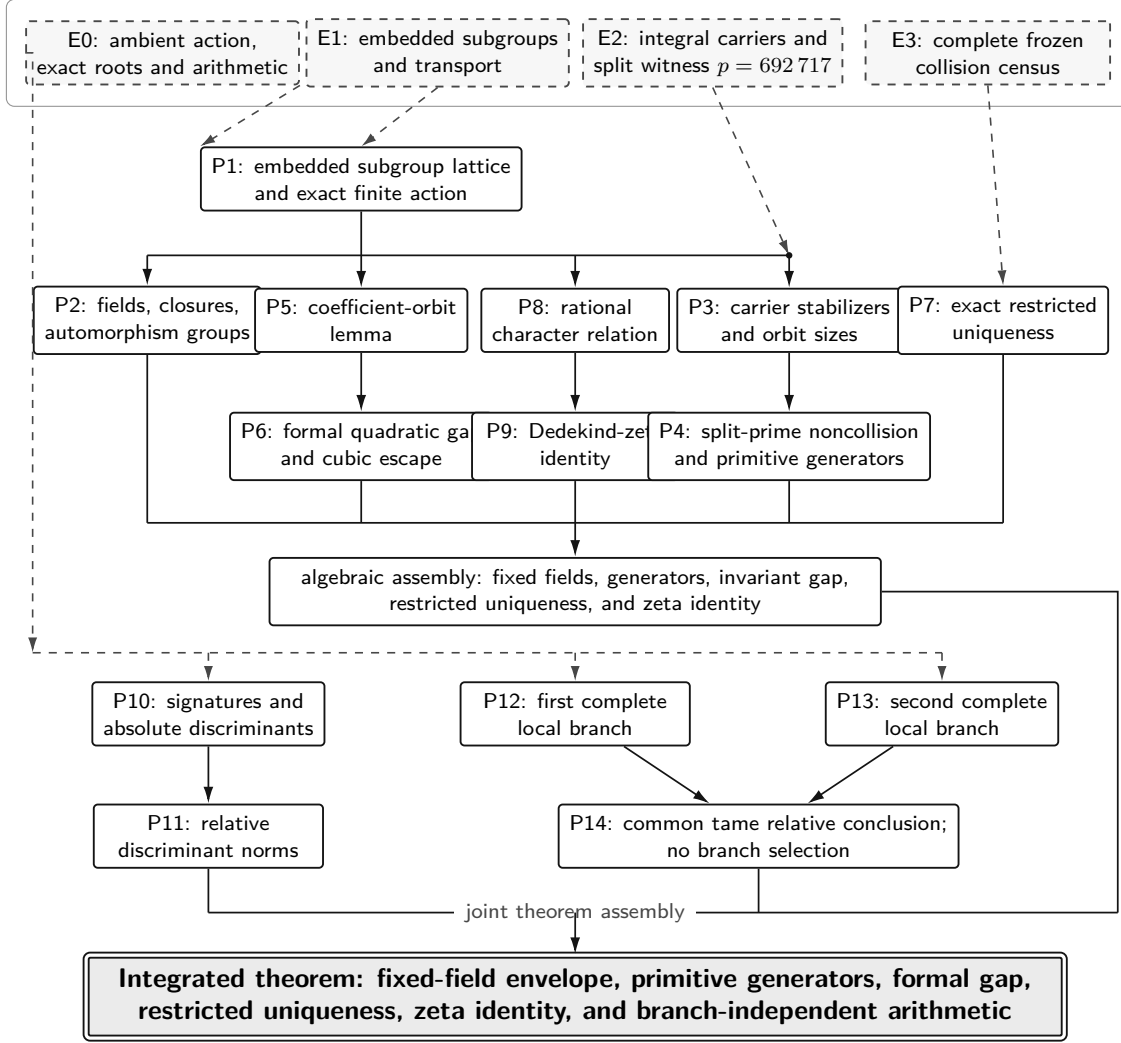


FIGURE 2. Dependency structure of the integrated theorem. Dashed edges bind exact finite inputs to mathematical propositions; solid edges are proof implications. Field definitions precede primitive generators and arithmetic, the restricted census claim is kept separate from the zeta bridge, and the two complete local branches meet only at the common tame conclusion.

The splitting field K of the labelled line carrier is normal and the faithful line action identifies

$$G = \text{Gal}(K/\mathbf{Q}) = W(E_6), \quad |G| = 51840. \quad (2.3)$$

The six one-line generators of this action, the line-label bridge, and the split specialization are recorded in the anonymous exact supplement. The surface-to-line realization is consistent with the general cubic-surface inverse-Galois framework of Elsenhans and Jahnel [2], but the particular coefficients in (2.1) and their certificate are inputs of this paper, not attributions to that work.

The normal closure has exact finite ramified support

$$\{3, 5, 181, 283, 997, 1801, 2346241, q\}, \quad (2.4)$$

$$q = 14932047182473291995860108491583652133938007263719.$$

At 3 the released filtration leaves exactly two decomposition-group possibilities,

$$D_3 = \text{ToM } 140 \quad \text{or} \quad D_3 = \text{ToM } 206, \quad I_3 = \text{ToM } 140 \quad (2.5)$$

in the fixed TomLib table. The locators in (2.5) are versioned checks; the embedded generator arrays and filtration inclusions in the supplement are the durable data. We retain both possibilities throughout [section 7](#).

2.2. The embedded Gassmann pair. The durable one-line generators in [appendix A](#) define two exact order-162 subgroups $H_+, H_- \leq G$. Their frozen TomLib locators are 301 and 303, but those integers do not define the subgroups. Their indices are 320, their cores in G are trivial, and their rational permutation characters are equal. Thus their fixed fields are arithmetically equivalent in the sense of the classical Gassmann–Perlis criterion [6]. All statements in this paper refer to these specific embeddings.

For later carrier definitions, let

$$\mathcal{S}_+ = H_+ \cdot \{1, 2\} \sqcup H_+ \cdot \{1, 9\}, \quad \mathcal{S}_- = H_- \cdot \{1, 2\}. \quad (2.6)$$

The two plus orbits have size 27 each and the minus orbit has size 81. The associated integral quadratic sums are

$$\eta_+ = \sum_{\{i,j\} \in \mathcal{S}_+} \alpha_i \alpha_j, \quad \eta_- = \sum_{\{i,j\} \in \mathcal{S}_-} \alpha_i \alpha_j. \quad (2.7)$$

Their formal support stabilizers are H_+ and H_- . The scaling in (2.2) is part of the definition; no unscaled sum is used in an integrality or modular-polynomial claim.

2.3. Left transport and fixed-field notation. One-line permutations in the mathematical exposition act on labels on the left. The fixed transport is

$$x = [1, 15, 14, 13, 22, 12, 27, 26, 25, 7, 24, 16, 17, 6, 19, 18, 5, 20, 4, 21, 3, 2, 11, 10, 9, 23, 8]. \quad (2.8)$$

Thus

$$H_3 = xH_-x^{-1}, \quad \mathcal{S}_3 = x\mathcal{S}_-, \quad \eta_3 = x\eta_-. \quad (2.9)$$

The same operation is written H_-^x in GAP’s right-labelled permutation syntax, because $(i^h)^x = (i^x)^{h^x}$. This is a conversion of notation, not an inverse choice. Exact enumeration gives

$$\text{Stab}_G(\mathcal{S}_3) = H_3 \subset N, \quad (2.10)$$

where N is defined intrinsically in [section 3](#). In particular, $K^{H_3} = x(K^{H_-})$ is a $\mathbf{Q}$ -conjugate of the original minus field. We will not identify the two embedded fields.

We use the following standard formulas repeatedly. If $H \leq G$, then the normal closure of $K^H/\mathbf{Q}$ inside K is fixed by

$$\text{Core}_G(H) = \bigcap_{g \in G} gHg^{-1},$$

and restriction induces

$$\text{Aut}_{\mathbf{Q}}(K^H) \cong N_G(H)/H. \quad (2.11)$$

When $J \triangleleft N$, the extension K^J/K^N is Galois with group N/J . These formulas turn the finite lattice of the next section into the field-theoretic assertions of [theorem 1.1](#).

3. THE COMMON NORMALIZER AND THE FIXED-FIELD ENVELOPE

The central group-theoretic feature is stronger than equality of two permutation characters: after the transport in (2.9), the two embedded subgroups share the same index-two normalizer. Define

$$N = N_G(H_+) = N_G(H_3), \quad (3.1)$$

let H_0 be the third index-two subgroup of N containing the common derived subgroup, and put

$$J = H_+ \cap H_3. \quad (3.2)$$

The one-line generators in [appendix A](#) make these definitions entirely independent of Table-of-Marks numbering.

Proposition 3.1 (Exact subgroup lattice). *The five subgroups have the following invariants.*

group	order	$[G : H]$	SmallGroup	ToM	Core $_G(H)$	$N_G(H)$
N	324	160	[324, 39]	327	1	N
H_+	162	320	[162, 11]	301	1	N
H_0	162	320	[162, 10]	302	1	N
H_3	162	320	[162, 19]	303	1	N
J	81	640	—	266	1	N

Moreover

$$[N, N] = J, \quad N^{\text{ab}} \cong C_2 \times C_2, \quad N/J \cong V_4, \quad (3.3)$$

and for distinct $i, j \in \{+, 0, 3\}$,

$$H_i \cap H_j = J, \quad \langle H_i, H_j \rangle = N. \quad (3.4)$$

Proof. Generate the subgroups from the one-based image arrays in [appendix A.1](#). Closure gives the displayed orders. Direct conjugation by every element of the 51840-element labelled action gives the normalizers and cores in the table. The commutator subgroup of N has order 81 and its element set equals the generated subgroup J ; the quotient has order four and SmallGroup identifier [4, 2], hence is V_4 . The three index-two subgroups are the inverse images of the three order-two subgroups of this quotient. Their pairwise intersections are therefore the kernel J , and the images of any two distinct ones generate V_4 , proving (3.4). The supplementary group certificate repeats these calculations from the durable arrays in two independent implementations. $\square$

Define the fixed fields

$$M = K^N, \quad F_i = K^{H_i} \ (i \in \{+, 0, 3\}), \quad L = K^J. \quad (3.5)$$

Corollary 3.2 (Biquadratic diamond). *The inclusions and degrees are*

$$\mathbf{Q} \subset M \subset F_i \subset L \subset K, \quad [M : \mathbf{Q}] = 160, \ [F_i : \mathbf{Q}] = 320, \ [L : \mathbf{Q}] = 640.$$

The extension L/M is Galois with group V_4 , and the three F_i are exactly its quadratic intermediate fields.

Proof. The degree equalities are $[K^H : \mathbf{Q}] = [G : H]$. Inclusions reverse subgroup inclusions, so $J \subset H_i \subset N$ gives $M \subset F_i \subset L$. Since $J \triangleleft N$, Galois correspondence over M identifies $\text{Gal}(L/M)$ with N/J . The three groups H_i/J are precisely the three order-two subgroups of $N/J \cong V_4$, proving the last assertion. $\square$

Proposition 3.3 (Normal closures and automorphisms). *Each of M, F_+, F_0, F_3, L has normal closure K over $\mathbf{Q}$, and*

$$\text{Aut}_{\mathbf{Q}}(M) = 1, \quad \text{Aut}_{\mathbf{Q}}(F_i) \cong C_2, \quad \text{Aut}_{\mathbf{Q}}(L) \cong V_4.$$

Furthermore $F_3 = x(K^{H_-})$.

Proof. The core of every defining subgroup is trivial by [proposition 3.1](#); hence the normal closure of each fixed field inside K is fixed by the trivial subgroup and equals K . Formula (2.11) and the normalizer column give

$$N_G(N)/N = 1, \quad N/H_i \cong C_2, \quad N/J \cong V_4.$$

Finally, x fixes $\mathbf{Q}$ and sends the H_- -fixed elements to the xH_-x^{-1} -fixed elements. This proves $x(K^{H_-}) = K^{H_3}$. It does not put the original embedded minus field inside this chosen L ; only its transported conjugate occurs in the diamond. $\square$

The lattice is now defined independently of any primitive element. This order of proof is deliberate: the carriers in the next section identify explicit generators for fields already determined by Galois correspondence, rather than defining fields retroactively from a resolvent degree.

4. PRIMITIVE GENERATORS AND THE FORMAL INVARIANT GAP

The Galois correspondence in [section 3](#) identifies the fields but does not exhibit elements that generate them. We now construct such elements from the integral line roots. The method is in the classical stabilizer–resolvent tradition [\[3, 7\]](#); the exact supports, the cubic obstruction, and the split witness are specific to the present labelled action.

4.1. Three formal carriers. Retain η_+ and $\eta_3 = x\eta_-$ from [equations \(2.7\) and \(2.9\)](#). Let τ be the exact element of $N \setminus H_+$ whose one-line array is displayed in [appendix B.1](#). Define

$$\mu = \eta_+ + \tau(\eta_+), \quad \lambda = \eta_+ + 2\eta_3. \quad (4.1)$$

The coefficient 2 in λ records the two support colors and prevents an element from interchanging them. The first carrier has 81 distinct quadratic monomials with coefficient histogram 54 of weight one and 27 of weight two. The second has 135 monomials, 54 of weight one and 81 of weight two.

For the companion field, let $\mathcal{T}_0$ be the following orbit of triples, written without compressed decimal notation:

$$\begin{aligned} \mathcal{T}_0 = \{ & \{1, 2, 9\}, \{1, 13, 18\}, \{1, 14, 25\}, \{2, 10, 14\}, \{2, 18, 26\}, \{3, 6, 12\}, \{3, 11, 23\}, \\ & \{3, 15, 19\}, \{4, 5, 21\}, \{4, 7, 24\}, \{4, 16, 17\}, \{5, 7, 20\}, \{5, 16, 27\}, \{6, 8, 11\}, \\ & \{6, 19, 22\}, \{7, 17, 27\}, \{8, 12, 15\}, \{8, 19, 23\}, \{9, 10, 13\}, \{9, 25, 26\}, \{10, 18, 25\}, \\ & \{11, 15, 22\}, \{12, 22, 23\}, \{13, 14, 26\}, \{16, 20, 24\}, \{17, 20, 21\}, \{21, 24, 27\} \}. \end{aligned} \quad (4.2)$$

The cubic carrier is

$$\xi_0 = \sum_{\{i,j,k\} \in \mathcal{T}_0} \alpha_i \alpha_j \alpha_k. \quad (4.3)$$

The complete set in (4.2) is the definition. We retain the redundant support hashes in [table 2](#) to catch transcription or index-origin errors.

4.2. Integrality and split-prime noncollision.

Lemma 4.1 (Integral scaling). *Each $\alpha_i = L_0 d_i$ is an algebraic integer. Hence μ, ξ_0, λ are algebraic integers.*

Proof. Substitute $d = y/L_0$ into $g(d) = L_0 d^{27} + b_{26} d^{26} + \cdots + b_0$ and multiply by L_0^{26} . The result is the monic integral equation

$$y^{27} + b_{26} y^{26} + L_0 b_{25} y^{25} + \cdots + L_0^{25} b_1 y + L_0^{26} b_0 = 0.$$

It is satisfied by $y = L_0 d_i$. Integer polynomials in algebraic integers are integral, which applies to all three finite sums. $\square$

The weighted-carrier stabilizers are exact formal statements:

$$\text{Stab}_G(\mu) = N, \quad \text{Stab}_G(\xi_0) = H_0, \quad \text{Stab}_G(\lambda) = J \quad (4.4)$$

when the symbols α_i are treated as independently labelled variables. This is an exhaustive assertion for all three carriers: for each of the 51840 elements of the durable labelled action, the canonical weighted monomial dictionary is transported and compared with the original. The elements passing that equality test form exactly N for μ , exactly H_0 for ξ_0 , and exactly J for λ , of orders 324, 162, 81. Thus the calculation excludes a larger ambient stabilizer; it does not merely show that the named subgroup fixes the carrier. Structurally, the two H_+ -support translates in μ are exchanged by N/H_+ ; unequal weights in λ recover $\mathcal{S}_+$ and $\mathcal{S}_3$ separately; and the 27 triples in ξ_0 form one H_0 -orbit. Formal stabilizers still do not exclude equalities introduced by evaluation at the actual roots. The next lemma supplies that bridge.

Lemma 4.2 (Split-prime noncollision). *Let θ be an integral polynomial in the labelled roots, fixed formally by $H \leq G$. Suppose that, at a common unramified prime where all labelled roots lie in the residue field, the $[G : H]$ values obtained from the formal G/H orbit of θ are pairwise distinct. Then*

$$\text{Stab}_G(\theta) = H, \quad \mathbf{Q}(\theta) = K^H.$$

Its orbit product is the monic integral irreducible separable minimal polynomial of degree $[G : H]$.

Proof. If two characteristic-zero integral conjugates were equal, their difference would be the zero algebraic integer and would reduce to zero at every common integral specialization. This contradicts distinctness of the stated residues. Thus the evaluated orbit has at least $[G : H]$ elements. Since H fixes θ , it has at most that many elements. Orbit-stabilizer therefore gives equality and the exact stabilizer H . Galois correspondence gives $\mathbf{Q}(\theta) = K^H$. The orbit product is fixed by G , has rational integral coefficients, and contains each conjugate once; hence it lies in $\mathbf{Z}[T]$ and is the minimal polynomial. Characteristic zero gives separability. $\square$

The common witness is

$$p = 692717. \quad (4.5)$$

It lies outside (2.4). The normalized eliminant is squarefree and factors as 1^{27} modulo p ; the 27 roots are distinct, the line equations vanish, and the reconstructed line graph carries the full 51840-element faithful action. The unramified Frobenius fixes all 27 labels, so faithfulness makes it the identity in G ; hence p splits completely in K and all orbit values reduce in the same residue field. Complete evaluation gives respectively 160, 320, and 640 distinct values. The exact product fingerprints are as follows.

Applying lemma 4.2 to the three rows proves

$$M = \mathbf{Q}(\mu), \quad F_0 = \mathbf{Q}(\xi_0), \quad L = \mathbf{Q}(\lambda). \quad (4.6)$$

The resulting orbit polynomials are exact in product form. We do not claim expanded characteristic-zero coefficient vectors or hashes for them.

4.3. Why the companion first appears in degree three. The cubic in (4.3) is not merely the first successful item in a coefficient search. Its degree is forced by a complete invariant-space argument in the formal coordinate ring.

Proposition 4.3 (Coefficient-orbit lemma). *Let $R = \mathbf{Q}[X_1, \dots, X_{27}]$ with G permuting the variables. If two subgroups $H \leq N \leq G$ have the same orbit partitions on points and unordered pairs, then*

$$R_{\leq 2}^H = R_{\leq 2}^N.$$

TABLE 2. Primitive-carrier certificate. Carrier hashes use canonical compact JSON with zero-based monomial labels; every mathematical display in the paper uses one-based labels. The last column hashes the complete modular coefficient vector, not an expanded characteristic-zero polynomial.

field	terms/degree	stabilizer/orbit	formal carrier SHA-256	modular coefficient SHA-256
M	81/2	324/160	0beb2791f4df4bb56214b6a35384517 083f5909004219cc988b6de70f494d1 7c	b8818888c1ceb83e05d2f2df045e9d6 e418f1ea18a5f019d1398e4cd0a59ef 6b
F_0	27/3	162/320	83f014bb3087708ad6e65c4f61bc92a 73172aa649ef573358164c1ae7d9efb c5	ffe9439cd390729bbb0dd7ffa4c6a10 45c7fbc9c645e0f37e75c71d1e786e1 0d
L	135/2	81/640	fae69eb91d414d8241bbbee51f4a3fc c91c4f8691090adc5cbb575079d2ea1 f5	c82feda40496156b7d006de4e47a1b8 08b3cf3fffe4a386652d3e3fa77861 f1

TABLE 3. Formal invariant-gap audit. The H_0 and N coefficient orbits agree on the constant, point/square, and unordered-pair bases, so every commutative formal invariant of degree at most two fixed by H_0 is fixed by N . This statement is made before imposing relations among the labelled roots. The cubic carrier ξ_0 has exact stabilizer H_0 and escapes the obstruction.

formal coefficient basis	H_0 orbit sizes	N orbit sizes	audit consequence
linear X_i and squares X_i^2	[27]	[27]	equal
mixed $X_i X_j$ ($i < j$)	[27, 27, 54, 81, 162]	[27, 27, 54, 81, 162]	equal
cubic carrier ξ_0	27 monomials; $\text{Stab}_G(\xi_0) = H_0$, $ H_0 = 162$		G -orbit 320

The equality concerns commutative $\mathbf{Q}$ -coefficient formal polynomials in independent labelled variables before quotienting by relations among the line roots. It excludes an exact H_0 stabilizer in degree at most two; the displayed cubic carrier escapes the obstruction.

In particular, no formal polynomial of total degree at most two can have stabilizer exactly H .

Proof. Every element of $R_{\leq 2}$ has a unique expansion in the monomial basis

$$1, \quad X_i, \quad X_i^2, \quad X_i X_j \quad (i < j).$$

An H -fixed polynomial has coefficients constant on H -orbits. The linear and square coefficients are indexed by points; the mixed coefficients are indexed by unordered pairs. If the two partitions agree with those for N , each coefficient function is also constant on N -orbits. Therefore every H -fixed polynomial of degree at most two is N -fixed. The reverse inclusion follows from $H \leq N$, proving equality. An element whose stabilizer is exactly H would be H -fixed but not N -fixed, so none exists. $\square$

For $H = H_0$, exact orbit enumeration gives the data in [table 3](#). The two partitions agree as sets, not merely as multisets of sizes. By [proposition 4.3](#), no commutative $\mathbf{Q}$ -coefficient formal polynomial in the labelled variables of total degree at most two has exact stabilizer H_0 . In contrast, the orbit sum (4.3) has exact stabilizer H_0 and is primitive by the split witness.

This conclusion is deliberately formal. Evaluating $X_i \mapsto \alpha_i$ may impose algebraic relations among the roots, and we do not classify expressions in the resulting quotient

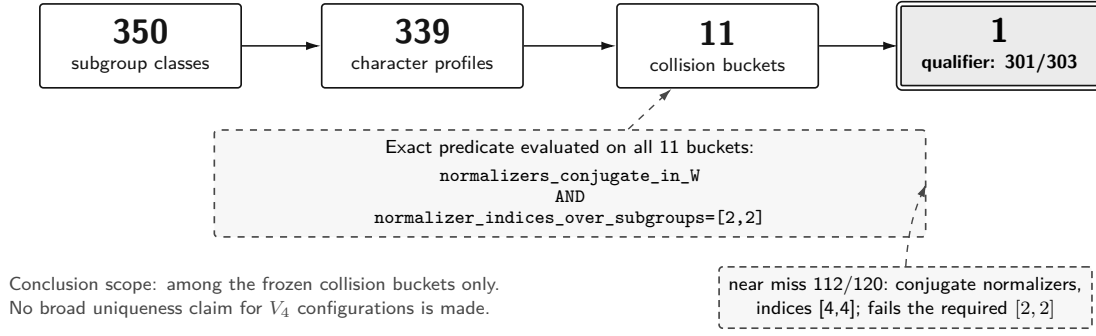


FIGURE 3. The exhaustive restricted-uniqueness funnel. The predicate `normalizers_conjugate_in_W` AND `normalizer_indices_over_subgroups=[2,2]` is evaluated without a target filter on every frozen collision bucket. Only 301/303 qualifies. The 112/120 bucket is essential: its normalizers are conjugate, but their indices $[4,4]$ fail the required index-two condition. No broader uniqueness of a V_4 configuration is asserted.

ring. The theorem says precisely that degree two cannot distinguish H_0 from N at the level of commutative formal coefficient orbits, while the displayed cubic does.

5. EXHAUSTIVE UNIQUENESS, CHARACTERS, AND ZETA FUNCTIONS

The common-normalizer construction sits inside a known finite collision census. We first delimit exactly what is unique, and then use the resulting V_4 lattice to obtain a representation-ring identity.

5.1. The exact eleven-bucket predicate. For an unordered collision bucket $\{a, b\}$ represented by embedded subgroups $H_a, H_b \leq G$, define the predicate

$$\begin{aligned} \mathcal{P}(a, b) : \quad & N_G(H_a) \text{ and } N_G(H_b) \text{ are conjugate in } G, \\ & [N_G(H_a) : H_a] = [N_G(H_b) : H_b] = 2. \end{aligned} \tag{5.1}$$

No degree or target locator is included in $\mathcal{P}$. The computation starts from all 350 subgroup conjugacy classes and their 339 distinct permutation-character profiles; it then evaluates (5.1) on every one of the eleven collision buckets recorded by James [4, §3.1, printed p. 22].

Proposition 5.1 (Restricted exhaustive uniqueness). *Among the eleven collision buckets, $\mathcal{P}(a, b)$ holds exactly for $\{a, b\} = \{301, 303\}$.*

Proof. For each bucket, generate both exact subgroups, compute their normalizers in the full 51840-element group, test normalizer conjugacy in G , and divide the two normalizer orders by the subgroup order. The complete rows are in table 6. Only the final row has conjugate normalizers and index pair $[2, 2]$. The calculation contains no target-filtering step: all eleven Boolean leaves are computed before the unique qualifying list is formed. $\square$

The exhaustive count is summarized in figure 3. This count does not say that the envelope is a globally unique V_4 configuration. The bucket 112/120 is the instructive near-miss: its normalizers are conjugate, and after transport the two order-16 groups generate a group of order 32 with intersection of order 8. Nevertheless each normalizer has order 64, so the normalizer indices are $[4, 4]$ and $\mathcal{P}(112, 120)$ is false. Dropping the index-two clause would therefore support a false uniqueness sentence.

5.2. The rational V_4 relation. Let $V = N/J \cong V_4$ and let A_+, A_0, A_3 be its three order-two subgroups, the images of H_+, H_0, H_3 . In the rational representation ring of V ,

$$[V/1] + 2[V/V] = [V/A_+] + [V/A_0] + [V/A_3]. \quad (5.2)$$

Indeed, the regular character is the sum of the trivial character and the three nontrivial one-dimensional characters. Each two-point permutation character $\mathbf{Q}[V/A_i]$ is the sum of the trivial character and the unique nontrivial character with kernel A_i . Both sides of (5.2) therefore contain the trivial character three times and each nontrivial character once.

Inflate (5.2) along $N \twoheadrightarrow N/J$ and induce from N to G . Transitivity of induction yields

$$[G/J] + 2[G/N] = [G/H_+] + [G/H_0] + [G/H_3]. \quad (5.3)$$

The independent computation in table 7 verifies (5.3) on all 25 conjugacy classes, with coefficient order $(H_+, H_0, H_3, J, N) = (-1, -1, -1, 1, 2)$. Checking all classes prevents a matching-dimensions argument from being mistaken for character equality.

Remark 5.2 (Characters, not finite sets). The two sides of (5.3) are equal after passing to rational permutation representations. They are not isomorphic finite G -sets: the orbit decompositions have different numbers and stabilizer types. Every use below is additive in characters.

5.3. Dedekind zeta consequence. For $H \leq G$, Artin formalism identifies the Dedekind zeta function of K^H with the Artin L -function attached to $\text{Ind}_H^G \mathbf{1}$. Multiplicativity on character sums, as in Perlis's criterion [6], converts (5.3) into

$$\zeta_L(s)\zeta_M(s)^2 = \zeta_{F_+}(s)\zeta_{F_0}(s)\zeta_{F_3}(s). \quad (5.4)$$

The original H_+, H_- pair has equal permutation characters, and conjugation by x preserves the character. Hence $\zeta_{F_3} = \zeta_{F_+}$ and

$$\zeta_L(s)\zeta_M(s)^2 = \zeta_{F_+}(s)^2\zeta_{F_0}(s). \quad (5.5)$$

The generic V_4 identity is classical; the contribution here is its realization by the exact embedded fields together with their explicit generators and arithmetic.

Equations (5.4)–(5.5) are the entire analytic conclusion. We compute no new decomposition Frobenius, bad Artin Euler factor, epsilon factor, or root number, and infer no holomorphy or automorphy statement. The next section uses conductor characters only to compute discriminants, a logically separate finite invariant.

6. SIGNATURES AND ABSOLUTE AND RELATIVE DISCRIMINANTS

The subgroup lattice gives five transitive coset characters. Evaluating those characters on complex conjugation and on the released ramification filtrations determines the signatures and conductor-discriminant exponents. This section gives the complete derivation; the prime-by-prime ledger is repeated in appendix C.

6.1. Archimedean signatures. Let $E = K^H$ have degree $n = [G : H]$, and let $o_c(H)$ be the number of orbits of a complex-conjugation subgroup on G/H . Real embeddings are fixed points and nonreal embeddings occur in transposed pairs, so

$$r_1 + 2r_2 = n, \quad r_1 + r_2 = o_c(H).$$

Therefore

$$r_1 = 2o_c(H) - n, \quad r_2 = n - o_c(H). \quad (6.1)$$

The exact pairs (n, o_c) are

$$(160, 88), \quad (320, 168), \quad (320, 160), \quad (640, 320)$$

for $M, F_+/F_3, F_0$, and L . Substitution in (6.1) gives

$$(16, 72), \quad (16, 152), \quad (0, 160), \quad (0, 320). \quad (6.2)$$

TABLE 4. Complete coset-orbit input for signatures and discriminants.
The orbit vector is ordered as $(I_3, P_3, Q_3, I_5, P_5, C_3, C_2, c_\infty)$.

field	orbit-count vector	(v_3, v_5, v_A, v_B)
M	$(22, 28, 56, 8, 32, 64, 80, 88)$	$(308, 248, 96, 80)$
F_+, F_3	$(36, 56, 112, 16, 64, 128, 160, 168)$	$(624, 496, 192, 160)$
F_0	$(28, 56, 112, 16, 64, 128, 160, 160)$	$(632, 496, 192, 160)$
L	$(56, 112, 224, 32, 128, 256, 320, 320)$	$(1264, 992, 384, 320)$

Each r_2 is even, so the sign $(-1)^{r_2}$ of every field discriminant is positive.

6.2. Conductor exponents. The released lower filtration uses the ordered subgroup types

$$(I_3, P_3, Q_3, I_5, P_5, C_3, C_2, c_\infty).$$

For a degree- n coset carrier, write o_U for the number of U -orbits. For a local lower filtration $G_0 \supseteq G_1 \supseteq \cdots$, the Artin conductor of the permutation character is

$$a = \sum_{u \geq 0} \frac{|G_u|}{|G_0|} (n - o_{G_u}). \quad (6.3)$$

At 3, the inertia group has order 18, the wild group has order 9, and the order-3 group Q_3 occurs in six positive lower grades. Their total weights in (6.3) are therefore 1, $9/18 = 1/2$, and $6(3/18) = 1$. At 5, inertia has order 20 and the order-5 wild group occurs in three positive grades, giving weights 1 and $3(5/20) = 3/4$. A tame cyclic inertia group contributes only grade zero. Consequently the conductor-discriminant formula gives

$$v_3 = (n - o_{I_3}) + \frac{1}{2}(n - o_{P_3}) + (n - o_{Q_3}), \quad (6.4)$$

$$v_5 = (n - o_{I_5}) + \frac{3}{4}(n - o_{P_5}), \quad (6.5)$$

$$v_A = n - o_{C_3}, \quad v_B = n - o_{C_2}. \quad (6.6)$$

Here v_A is the common exponent at the three tame C_3 primes 181, 997, 2346241, and v_B is the common exponent at the three reflection C_2 primes 283, 1801, q .

The orbit counts and resulting exponents appear in table 4. For example, the exponent of 3 for M is

$$(160 - 22) + \frac{1}{2}(160 - 28) + (160 - 56) = 308,$$

and the other entries follow by the same substitutions. The complete character values rather than only the displayed totals are bound in the exact certificate.

Define

$$q = 14932047182473291995860108491583652133938007263719, \quad (6.7)$$

$$A = 181 \cdot 997 \cdot 2346241 = 423395612137, \quad (6.8)$$

$$B = 283 \cdot 1801 \cdot q \quad (6.9)$$

$$= 7610610604104534884325967676315830570581925356194091077. \quad (6.10)$$

Proposition 6.1 (Signed absolute discriminants). *The signed field discriminants are*

$$\begin{aligned} \text{Disc}(M) &= +3^{308}5^{248}A^{96}B^{80}, \\ \text{Disc}(F_+) = \text{Disc}(F_3) &= +3^{624}5^{496}A^{192}B^{160}, \\ \text{Disc}(F_0) &= +3^{632}5^{496}A^{192}B^{160}, \\ \text{Disc}(L) &= +3^{1264}5^{992}A^{384}B^{320}. \end{aligned}$$

Each field has exact finite ramified support (2.4).

Proof. Equations (6.4)–(6.6) and table 4 give the prime exponents. The even values of r_2 in (6.2) give the positive signs. The normal closure is unramified outside (2.4), so each fixed field is also unramified outside it. Conversely, every exponent in each displayed factorization is positive; after expanding A and B , all eight primes occur. This proves exactness of the support as well as the factorizations. $\square$

6.3. Relative discriminants over the common core field. For a finite extension E/M , the discriminant tower identity is

$$\text{Disc}(E/\mathbf{Q}) = \text{Disc}(M/\mathbf{Q})^{[E:M]} N_{M/\mathbf{Q}}(\mathfrak{d}_{E/M}). \quad (6.11)$$

Subtracting twice the exponent vector of M from the three quadratic rows gives

$$\begin{aligned} F_+/M &: (624, 496, 192, 160) - 2(308, 248, 96, 80) = (8, 0, 0, 0), \\ F_0/M &: (632, 496, 192, 160) - 2(308, 248, 96, 80) = (16, 0, 0, 0), \\ F_3/M &: (624, 496, 192, 160) - 2(308, 248, 96, 80) = (8, 0, 0, 0). \end{aligned}$$

For the degree-four top,

$$(1264, 992, 384, 320) - 4(308, 248, 96, 80) = (32, 0, 0, 0).$$

We have proved

$$\begin{aligned} N_{M/\mathbf{Q}} \mathfrak{d}_{F_+/M} &= 3^8, & N_{M/\mathbf{Q}} \mathfrak{d}_{F_0/M} &= 3^{16}, \\ N_{M/\mathbf{Q}} \mathfrak{d}_{F_3/M} &= 3^8, & N_{M/\mathbf{Q}} \mathfrak{d}_{L/M} &= 3^{32}. \end{aligned} \quad (6.12)$$

In particular, all four relative extensions are unramified away from primes of M above 3. The absolute identity alone does not specify the local factorization above 3; that information comes from the complete double-coset tables in the next section.

7. BOTH RETAINED 3-ADIC TOWERS

The global calculation leaves two possible decomposition groups at 3. Rather than select one, we compute the complete relative tower in each case and isolate the conclusion common to both.

7.1. Double cosets and row conventions. Fix a prime of K above 3, with decomposition group D , inertia group I , and lower ramification filtration. For $F = K^H$, primes of F above 3 are indexed by double cosets

$$D \backslash G / H. \quad (7.1)$$

For a double-coset representative, intersections of its conjugate of H with D , I , and the lower filtration determine the local degree, ramification index, residue degree, and different exponent. We record an absolute factor by (n, e, f, d) , where $n = ef$ is its local degree and d is the exponent of its different. Over a fixed factor of M , a factor in one of F_+, F_0, F_3, L is recorded by (g, e, f, d) , where g is the number of relative factors over that fixed base prime and e, f, d are their common ramification index, residue degree, and different exponent. A multiplicity after a row counts base primes of M with the same complete relative pattern.

For every base prime and every relative field E , the rows satisfy

$$gef = [E : M]. \quad (7.2)$$

Summing local different contributions recovers the exponent of 3 in (6.12). These degree and different checks are performed row by row, not only after taking global totals.

TABLE 5. Comparison of both retained local branches. Factor and relative norm-exponent tuples are ordered as (F_+, F_0, F_3, L) . Every ramified relative row has $e = 2$ and $d = 1$ in both complete branches, while neither branch is selected.

ToM locator	base rows	factor totals (F_+, F_0, F_3, L)	v_3 of relative norms	ramified relative rows	branch status
140	22	(36,28,36,56)	(8,16,8,32)	$e = 2, d = 1$	not selected
206	11	(18,14,18,28)	(8,16,8,32)	$e = 2, d = 1$	not selected

The identical exponent tuple is ordered as (F_+, F_0, F_3, L) . Both complete decompositions imply the same tame relative ramification conclusion; the table does not select either decomposition-group branch.

7.2. Comparison of the two branches. The complete collected rows appear in [tables 12](#) and [13](#). Their principal totals are compared in [table 5](#).

The difference between 22 and 11 base primes reflects the residue-degree two contribution in the second decomposition group: the collected base rows for ToM 206 have residue degree 2, whereas those for ToM 140 have residue degree 1. This changes the number of factors but not the relative different exponents.

Proposition 7.1 (Branch-independent relative ramification). *For each of the two possibilities in (2.5), every relative extension $F_+/M, F_0/M, F_3/M, L/M$ is unramified outside primes above 3. Above 3, every ramified relative factor has $e = 2$ and $d = 1$. Consequently all relative ramification is tame, and the relative discriminant norms are those in (6.12) in both branches.*

Proof. Unramifiedness away from 3 follows already from the relative discriminant exponent differences in [section 6](#). At 3, enumerate (7.1) separately for the exact embedded filtrations of ToM 140 and ToM 206. The resulting collected rows are precisely [tables 12](#) and [13](#). In each relative entry, gef equals 2 for each quadratic field and 4 for L , as required by (7.2). Multiplicity-weighted factor counts give the middle column of [table 5](#); summing the relative different contributions gives (8, 16, 8, 32) in each branch.

Inspection of all relative entries shows that every ramified entry is either $(1, 2, 1, 1)$ or $(2, 2, 1, 1)$. Since the residue characteristic is 3 and 2 is prime to 3, such ramification is tame; for a tame extension the different exponent is $e - 1 = 1$, agreeing with the table. No other ramified row occurs. $\square$

The proposition is one-way. We do not infer that the actual decomposition group is 140 or 206, and a tuple (n, e, f, d) does not classify an individual high-degree local field up to isomorphism. In particular, the tables do not prove a converse local classification. Their role is to show that both retained global possibilities imply the same exact relative ramification statement.

8. EXACT COMPUTATION, REPRODUCIBILITY, AND SCOPE

The proof uses finite computation at four bounded points: subgroup closure and the collision census, stabilizers and modular orbit values, permutation characters and conductor orbits, and local double cosets. This section records how those premises are reconstructed and how they connect to the mathematical arguments. Test counts are evidence about implementation integrity, not substitutes for the proofs in [sections 3](#) to [7](#).

8.1. Proof-to-gate map. The anonymous certificate has eight gates.

gate	finite obligation	mathematical destination
G0	released normal extension, labelled lines, prior subgroup and local data	premises of section 2
G1	durable groups, transport, lattice, all eleven predicate rows	propositions 3.1 and 5.1
G2	three carriers, stabilizers, 160/320/640 mod- ular noncollision	lemma 4.2 and equation (4.6)
G3	exact point/pair partitions and cubic stabi- lizer	proposition 4.3
G4	cores, normalizers, all 25 character values, V_4 relation	proposition 3.3 and equation (5.3)
G5	archimedean and filtration orbit counts, exact support	proposition 6.1 and equation (6.12)
G6	every absolute and relative row for both D_3 branches	proposition 7.1
G7	independent reconstruction, schema, muta- tion, source, and scope controls	authority for the finite premises and the boundary below

The finite group calculations use GAP 4.11.1 and TomLib 1.2.9 [5, 8]; the frozen environment also records CTblLib 1.3.1 and SmallGrp 1.4.1. Package locators are always checked against durable permutation arrays. Thus a change in table ordering cannot silently redefine a field.

8.2. Independent reconstruction and adversarial checks. The paper-facing package contains 13 source files and 8 result files; the complete live inventory has 21 files, while the self-excluding scoped manifest binds 20. Two complete official cycles each pass 53/53 tests and reconstruct all eight gates. The canonical payload has 9310 scalar leaves. The independent checker rejects 9339 value mutations, 9339 type mutations, and 14 structural mutations. Its evidence-rebound suite rejects 6 group-document and 4 resolver-document mutations; two additional self-consistent artifact-row digest attacks give the explicit count summary 6/4/10/2/12. Thirty-nine child-input snapshots are rebound before the certificate write.

The group and primitive-resolver evidence are produced by separate lanes. The integrated checker does not call the producer’s theorem functions: it runs the final group checker to recover the canonical projection, invokes an independently implemented resolver reconstruction, and rebuilds all payload leaves. Strict parsers reject duplicate keys, booleans disguised as integers, unknown fields, symlinks, path substitution, stale caches, and input changes between validation and write. The exact schema, certificate, check report, component evidence, and self-excluding manifest are bound by the hashes in [table 14](#). These controls make the finite claims reproducible from the anonymous supplement while leaving the conceptual proof visible in the paper.

8.3. Scope firewall.

Scope. The exact uniqueness theorem is restricted to the eleven collision buckets and the predicate “normalizers conjugate in $W(E_6)$ and both normalizer indices equal two.” Orbit polynomials are exact in product form; no expanded characteristic-zero coefficient vector is claimed. The degree obstruction concerns commutative formal polynomials in independent labelled variables. The character relation is not a finite-set isomorphism. The zeta identity yields no new bad Euler factor, epsilon factor, root number, holomorphy, automorphy, or analytic-continuation statement. Neither D_3 branch is selected, and local tuples are not a classification of local fields. No integral-equivalence, integral-basis, maximal-order, class-number, regulator, trace-form, rational-point, motive, Hénnon, Yukawa, Calabi–Yau, dynamical, spectral, Riemann-hypothesis, or Hilbert–Pólya conclusion is made.

The literal scope tag in the certificate is

NO_BAD_EULER_OR_ROOT_NUMBER.

The full set of thirty false scope leaves is grouped and listed in [appendix D.3](#). This redundancy is intentional: a downstream use of a discriminant or zeta identity should not be able to erase the nonclaims that accompanied it.

8.4. Anonymous supplementary boundary. An anonymized supplementary archive is intended to contain canonical JSON, strict schemas, immutable manifests, durable group arrays, and replay instructions. The draft contains no repository URL, host name, user name, institutional path, commit identity, or identifying acknowledgement. The supplementary certificate’s machine-layer field `release_authorized=false` records that its finite evidence is not, by itself, a promotion or publication claim. That workflow field is not a mathematical premise of the theorem. Release and archival state are managed outside the manuscript so that the mathematical source remains stable across later packaging steps.

9. CONCLUSION

The exact common normalizer upgrades one pairwise Gassmann collision into a coherent fixed-field envelope. Its subgroup lattice gives a biquadratic degree-160/320/640 diamond; exact line-coordinate carriers make three levels primitive; equality of point and pair partitions explains why the new quadratic companion first appears through a cubic formal invariant; and the V_4 character relation organizes both the zeta identity and the relative arithmetic. Global conductor calculations and two separate local double-coset tables then show that all relative ramification is tame above 3, independently of which retained decomposition-group branch occurs.

Two natural questions remain outside the theorem. One is to resolve which of the two 3-adic decomposition groups belongs to the fixed normal extension. The other is to study integral refinements of the five fields beyond their discriminants. Neither question is answered by the present zeta or local-tuple data, and neither is needed for the envelope constructed here.

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APPENDIX A. DURABLE GROUP DATA AND THE COLLISION CENSUS

This appendix records the finite data needed to reproduce [propositions 3.1](#) and [5.1](#) and [equation \(5.3\)](#). All arrays are one-line images of $1, \dots, 27$ and are one-based. The Table-of-Marks integers elsewhere in the paper are versioned checks only.

A.1. Durable subgroup arrays. The ambient labelled $W(E_6)$ action is generated by the six simple-reflection arrays

G:

[2, 1, 3, 4, 5, 6, 7, 12, 13, 14, 15, 8, 9, 10, 11, 16, 17, 18, 19, 20, 21, 23, 22, 24, 25, 26, 27]
 [1, 3, 2, 4, 5, 6, 8, 7, 9, 10, 11, 12, 16, 17, 18, 13, 14, 15, 19, 20, 21, 22, 24, 23, 25, 26, 27]
 [1, 2, 4, 3, 5, 6, 7, 9, 8, 10, 11, 13, 12, 14, 15, 16, 19, 20, 17, 18, 21, 22, 23, 25, 24, 26, 27]
 [1, 2, 3, 5, 4, 6, 7, 8, 10, 9, 11, 12, 14, 13, 15, 17, 16, 18, 19, 21, 20, 22, 23, 24, 26, 25, 27]
 [1, 2, 3, 4, 6, 5, 7, 8, 9, 11, 10, 12, 13, 15, 14, 16, 18, 17, 20, 19, 21, 22, 23, 24, 25, 27, 26]
 [12, 8, 7, 4, 5, 6, 3, 2, 9, 10, 11, 1, 13, 14, 15, 16, 17, 18, 27, 26, 25, 22, 23, 24, 21, 20, 19]

They generate 51840 distinct permutations. The exact plus and original minus groups are generated by

H+:

[1, 2, 19, 21, 20, 3, 24, 11, 9, 10, 23, 15, 13, 14, 22, 5, 4, 18, 6, 16, 17, 12, 8, 27, 25, 26, 7]
 [16, 27, 13, 12, 22, 26, 15, 25, 24, 7, 14, 18, 20, 5, 1, 23, 8, 17, 9, 19, 6, 2, 10, 3, 4, 21, 11]
 [26, 13, 22, 20, 24, 15, 21, 3, 14, 1, 19, 11, 25, 18, 23, 7, 5, 9, 12, 27, 16, 8, 6, 17, 2, 10, 4]

H-:

[5, 1, 6, 2, 3, 4, 10, 21, 14, 17, 19, 11, 7, 8, 9, 15, 18, 20, 12, 13, 16, 26, 22, 27, 23, 24, 25]
 [7, 15, 13, 12, 26, 5, 16, 18, 20, 1, 22, 8, 9, 6, 27, 11, 4, 25, 3, 24, 14, 10, 21, 19, 17, 23, 2]
 [16, 23, 9, 8, 26, 27, 7, 25, 24, 10, 11, 12, 13, 6, 5, 22, 17, 18, 19, 20, 2, 1, 21, 3, 4, 15, 14]

The transported subgroup is $H_3 = xH_-x^{-1}$ with x in (2.8). The common normalizer is generated by

N:

[1, 18, 22, 16, 17, 12, 27, 8, 25, 26, 23, 6, 14, 13, 19, 4, 5, 2, 15, 21, 20, 3, 11, 24, 9, 10, 7]
 [15, 12, 4, 14, 25, 7, 10, 27, 8, 6, 17, 24, 11, 3, 21, 26, 13, 22, 5, 18, 1, 20, 16, 2, 19, 23, 9]
 [1, 2, 19, 21, 20, 3, 24, 11, 9, 10, 23, 15, 13, 14, 22, 5, 4, 18, 6, 16, 17, 12, 8, 27, 25, 26, 7]
 [14, 10, 3, 7, 4, 6, 5, 8, 2, 9, 11, 12, 1, 13, 15, 17, 27, 25, 19, 21, 24, 22, 23, 20, 26, 18, 16]
 [18, 1, 15, 4, 5, 12, 7, 6, 13, 14, 3, 8, 26, 25, 11, 16, 17, 2, 22, 20, 21, 23, 19, 24, 10, 9, 27]
 [1, 13, 16, 12, 6, 5, 8, 7, 9, 26, 27, 4, 2, 18, 17, 3, 15, 14, 20, 19, 22, 21, 24, 23, 25, 10, 11]

The third index-two subgroup and the common intersection are generated by

H0:

[2, 18, 23, 21, 20, 11, 24, 15, 26, 25, 22, 3, 9, 10, 19, 5, 4, 1, 8, 16, 17, 6, 12, 27, 14, 13, 7]
 [13, 9, 6, 16, 27, 19, 17, 23, 10, 2, 8, 22, 14, 1, 12, 24, 20, 26, 3, 7, 5, 15, 11, 4, 18, 25, 21]
 [15, 12, 4, 14, 25, 7, 10, 27, 8, 6, 17, 24, 11, 3, 21, 26, 13, 22, 5, 18, 1, 20, 16, 2, 19, 23, 9]
 [1, 2, 19, 21, 20, 3, 24, 11, 9, 10, 23, 15, 13, 14, 22, 5, 4, 18, 6, 16, 17, 12, 8, 27, 25, 26, 7]
 [20, 5, 3, 2, 9, 11, 10, 8, 7, 4, 6, 23, 24, 21, 19, 26, 18, 16, 15, 13, 1, 22, 12, 14, 17, 27, 25]

J:

[1, 2, 6, 17, 16, 19, 27, 23, 9, 10, 8, 22, 13, 14, 12, 20, 21, 18, 3, 5, 4, 15, 11, 7, 25, 26, 24]

[2, 18, 11, 4, 5, 8, 7, 12, 26, 25, 15, 6, 9, 10, 3, 16, 17, 1, 23, 20, 21, 19, 22, 24, 14, 13, 27]
 [3, 6, 5, 2, 1, 4, 18, 17, 12, 8, 16, 21, 15, 11, 20, 14, 10, 19, 7, 13, 9, 24, 27, 26, 23, 22, 25]

Generating from these arrays gives the invariant table in [proposition 3.1](#); element-set comparisons give all three meet/join identities. The exact transport check verifies $\text{Stab}_G(x\mathcal{S}_-) = xH_-x^{-1} \subset N$.

A.2. All eleven collision rows. For the near-miss 112/120, transport gives generated-group order 32 and intersection order 8 (frozen intersection locator 54). For 301/303 the corresponding values are 324 and 81 (intersection locator 266). These auxiliary facts explain why an unqualified “unique V_4 ” claim would be wrong; neither is part of the predicate.

A.3. All character values. The next table gives the complete vectors in the common conjugacy-class order. The first numerical column is the class size. A row-by-row check gives $\chi_J + 2\chi_N - \chi_+ - \chi_0 - \chi_3 = 0$.

TABLE 7. The 25 class values of the five transitive permutation characters.

class	size	χ_+	χ_0	χ_3	χ_J	χ_N
1	1	320	320	320	640	160
2	45	0	64	0	0	32
3	270	0	0	0	0	0
4	80	32	32	32	64	16
5	240	8	8	8	16	4
6	480	20	20	20	40	10
7	540	0	0	0	0	0
8	3240	0	0	0	0	0
9	5184	0	0	0	0	0
10	720	0	16	0	0	8
11	1440	0	4	0	0	2
12	1440	0	4	0	0	2
13	2160	0	0	0	0	0
14	5760	2	2	2	4	1
15	4320	0	0	0	0	0
16	36	0	0	0	0	0
17	540	16	0	16	0	16
18	540	0	0	0	0	0
19	1620	0	0	0	0	0
20	1440	0	0	0	0	0
21	1440	0	0	0	0	0
22	4320	4	0	4	0	4
23	6480	0	0	0	0	0
24	5184	0	0	0	0	0
25	4320	0	0	0	0	0

The equality $\chi_+ = \chi_3$ is visible in every row, while $\chi_+ \neq \chi_0$ at classes 2, 10, 11, 12, 17, 22. This supplies both the transported Gassmann equality and an independent check that the third quadratic companion is not being conflated with the pair.

APPENDIX B. PRIMITIVE-CARRIER CERTIFICATES

This appendix makes the definitions in [section 4](#) reproducible without expanding the degree-160, 320, and 640 characteristic-zero polynomials. Pair and triple labels in the paper are one-based; certificate hashes serialize the corresponding labels after subtracting one.

B.1. Exact supports and convention conversion. The element $\tau \in N \setminus H_+$ used in (4.1) has one-line array

[1, 13, 5, 15, 3, 20, 11, 24, 9, 26, 7, 21, 2, 18, 4, 19, 22, 14, 16, 6, 12, 17, 27, 8, 25, 10, 23]

It is the lexicographically least zero-based tuple in the exact set $N \setminus H_+$ (equivalently the displayed one-based image array after conversion). The plus support is

{1, 2} {1, 9} {1, 18} {1, 25} {2, 14} {2, 18} {2, 26} {3, 6} {3, 12}
 {3, 19} {3, 23} {4, 5} {4, 7} {4, 16} {4, 24} {5, 7} {5, 21} {5, 27}
 {6, 11} {6, 19} {6, 22} {7, 17} {7, 20} {8, 11} {8, 15} {8, 19} {8, 23}
 {9, 13} {9, 25} {9, 26} {10, 13} {10, 14} {10, 18} {10, 25} {11, 22}
 {11, 23} {12, 15} {12, 22} {12, 23} {13, 18} {13, 26} {14, 25} {14, 26}
 {15, 19} {15, 22} {16, 17} {16, 24} {16, 27} {17, 20} {17, 27} {20, 21}
 {20, 24} {21, 24} {21, 27}

It is the disjoint union of the two H_+ -orbits of $\{1, 2\}$ and $\{1, 9\}$. The transported minus support $\mathcal{S}_3 = x\mathcal{S}_-$ is

{1, 4} {1, 12} {1, 15} {1, 17} {1, 21} {1, 22} {2, 3} {2, 4} {2, 6}
 {2, 17} {2, 19} {2, 21} {3, 9} {3, 10} {3, 20} {3, 21} {3, 24} {4, 6}
 {4, 8} {4, 12} {4, 18} {5, 6} {5, 8} {5, 9} {5, 12} {5, 13} {5, 26}
 {6, 7} {6, 9} {6, 10} {7, 8} {7, 10} {7, 12} {7, 14} {7, 25} {8, 18}
 {8, 25} {8, 26} {9, 16} {9, 19} {9, 20} {10, 19} {10, 24} {10, 27}
 {11, 18} {11, 20} {11, 21} {11, 24} {11, 25} {11, 26} {12, 13}
 {12, 14} {13, 15} {13, 16} {13, 20} {13, 22} {14, 15} {14, 22}
 {14, 24} {14, 27} {15, 20} {15, 21} {15, 24} {16, 19} {16, 22}
 {16, 23} {16, 26} {17, 18} {17, 19} {17, 22} {17, 23} {18, 21}
 {18, 23} {19, 27} {20, 26} {22, 27} {23, 25} {23, 26} {23, 27}
 {24, 25} {25, 27}

These arrays and (4.2) give the carriers exactly:

$$\mu = \sum_{e \in \mathcal{S}_+} \alpha_e + \sum_{e \in \tau\mathcal{S}_+} \alpha_e, \quad \lambda = \sum_{e \in \mathcal{S}_+} \alpha_e + 2 \sum_{e \in \mathcal{S}_3} \alpha_e,$$

where $\alpha_{\{i,j\}}$ abbreviates $\alpha_i\alpha_j$, and $\xi_0 = \sum_{T \in \mathcal{T}_0} \prod_{i \in T} \alpha_i$. The intersection $\mathcal{S}_+ \cap \tau\mathcal{S}_+$ has 27 elements, explaining the weight histogram $54 \cdot 1 + 27 \cdot 2$ for μ ; $\mathcal{S}_+$ and $\mathcal{S}_3$ are disjoint, giving $54 \cdot 1 + 81 \cdot 2$ for λ .

Under the left action, applying x to every label of $\mathcal{S}_-$ gives $\mathcal{S}_3$ and $\text{Stab}_G(\mathcal{S}_3) = xH_-x^{-1} = H_3$. The right-labelled GAP notation for the same transported subgroup is H_-^x . Thus no inversion is introduced when comparing the mathematical and computational carriers.

B.2. The split specialization. Modulo $p = 692717$, the normalized line eliminant has sorted roots

[27847, 37592, 103493, 113820, 181112, 195162, 202320, 215888, 247842,
 261548, 267420, 275250, 305666, 312803, 355626, 392650, 430836, 477327,
 515776, 526394, 534416, 576186, 616498, 619023, 621746, 656335, 677763]

with SHA-256

96c072b9f1cc6a3e07d28433ed4b864b50f7d6841ed84a6536fd43ee050017dd.

After the certified actual-to-standard line relabelling, the scaled values α_i are

[437040, 472841, 94837, 293118, 84622, 361068, 136107, 24581, 223416,
 158756, 185411, 281614, 650803, 42897, 270349, 571692, 171614, 36060,
 254160, 327208, 483833, 567614, 384448, 7311, 183251, 111695, 484490].

All 27 roots are distinct; the factor pattern is 1^{27} ; every one of the four line equations vanishes for every reconstructed line; and the line intersection graph is carried isomorphically to the standard labelled graph. The actual-to-standard map has hash

a986674a4a0e64dc54244b81a491b86b4fe0af252fd5ad86512832641a64b0ea.

These facts make the specialization a completely split, unramified labelled witness rather than an unlabelled list of roots.

Pairwise distinctness of the values in each row proves characteristic-zero noncollision by [lemma 4.2](#). The argument needs only one good specialization: an equality of integral conjugates would survive every reduction, so one reduction at which all values separate rules it out.

APPENDIX C. GLOBAL AND LOCAL ARITHMETIC LEDGERS

C.1. Prime-by-prime discriminant data. Expanding A and B , [table 9](#) records the complete absolute exponent table. Together with the complex-conjugation orbit counts 88, 168, 160, 168, 320, this table reproduces all rows of [table 1](#). Subtracting two or four copies of the M row gives the relative vectors $(8, 0, \dots, 0)$, $(16, 0, \dots, 0)$, $(8, 0, \dots, 0)$, and $(32, 0, \dots, 0)$.

C.2. Absolute local summaries at 3. For compactness, r^m below means that the absolute row $r = (n, e, f, d)$ occurs with multiplicity m . The factor count, degree sum, and different sum follow each field.

C.3. Complete relative rows. The field order in each relative entry is (F_+, F_0, F_3, L) . A base row is an absolute (n, e, f, d) row of M ; each following entry is a relative (g, e, f, d) row, where g is the number of factors over the fixed base prime. The final multiplicity counts base primes of that pattern.

In every relative table entry, gef is 2 for a quadratic intermediate and 4 for the top. Summing $mf_M gfd$, where m is the displayed multiplicity and f_M is the residue degree in the base row, gives the four relative different totals 8, 16, 8, 32 in either table. The factor totals are obtained by summing g with multiplicity and equal $(36, 28, 36, 56)$ and $(18, 14, 18, 28)$, respectively.

APPENDIX D. EVIDENCE, SOURCE, AND SCOPE LEDGER

D.1. Claim-to-artifact binding. A strict schema validates every fixed basename in the anonymous supplement. The table binds the principal literal file bytes and the post-machine formal aggregate.

For completeness, the canonical payload SHA-256 is dca8dbbf269735e78b0435799b0d9c8c9ffad8bdd0470b9262ef64005ff0dead. The remaining stability bindings are the source-contract hash 4c484b3532c4604b028f45fc157c261149a7a49ca9631bbcf83f8d1efd1cdb90, the released-input authority-projection hash 0512db556004edde7c19176bbb35375beaeba89301da53902d5c5d98001cb8a8, and the official two-cycle transcript hash 5f5d788a1493c16a8eec86ec0cb40bfed2dea72fa2257bddf50eed1be2c43239.

From the root of the anonymized archive, the fixed wrapper code/run_all.sh builds a fresh canonical stage, runs the producer, the independent checker, the full test suite, and a nonmutating replay. It accepts no arbitrary repository path. Only the fixed code/results inventory is eligible for the self-excluding manifest, and all ordinary files are required to have one link and the declared mode. Reproducibility therefore binds content, schema, path identity, and the before/after state of every input.

D.2. Bounded source statement. The literature roles used here are intentionally narrow. Perlis supplies the character/zeta bridge [6]; Bosma–de Smit supply arithmetic-equivalence context [1]; James supplies the complete 350/339/11 census [4]; Stauduhar and Fieker–Klüners supply the classical resolvent and invariant framework [3, 7]; and Elsenhans–Jahnel supply broad cubic-surface realization context [2]. None of these sources is cited for the specific common normalizer, carriers, field diamond, or relative arithmetic. The bounded search ending 2026-08-16 found no source containing that exact conjunction. This negative search record supports the stated distinction from prior work; it is not a proof of absolute priority and is not converted into a “first” claim.

D.3. Thirty explicit false scope leaves. The machine certificate records the following thirty Boolean leaves as false. Their names are reproduced to prevent a prose summary from silently weakening the boundary. Workflow-named leaves state only what the frozen machine certificate itself does not claim; they are not a statement of the manuscript’s later archival state.

- (1) *Analytic and representation overreach:* artin_holomorphy_claimed, automorphy_claimed, bad_artin_euler_claimed, decomposition_frobenius_claimed, global_root_number_claimed, local_epsilon_factor_claimed, local_root_number_claimed, and finite_g_sets_isomorphic_from_character_relation.
- (2) *Integral and arithmetic overreach:* characteristic_zero_coefficient_hash_claimed, expanded_characteristic_zero_resolvent_claimed, class_number_claimed, integral_basis_claimed, maximal_order_claimed, monogenicity_claimed, regulator_claimed, and trace_form_claimed.
- (3) *Geometric and speculative overreach:* brauer_manin_claimed, hasse_principle_claimed, weak_approximation_claimed, rational_point_claimed, motive_claimed, rh_claimed, and hilbert_polya_operator_claimed.
- (4) *Formal, local, and authority overreach:* d3_branch_selected, formal_invariant_statement_after_root_relations, local_fields_classified_by_nefd_rows, raw_tom_defines_fields, target_selection_pilot_is_theorem_authority, paper_complete_claimed, and release_claimed.

The list contains $8 + 8 + 7 + 7 = 30$ leaves. In particular, every calculation in [sections 6 and 7](#) stays inside the literal firewall NO_BAD_EULER_OR_ROOT_NUMBER.

TABLE 6. All frozen collision buckets used by the exact predicate. The near miss has conjugate normalizers but indices different from the required pair; only the final row satisfies both predicate clauses.

bucket	$ H $	$[G : H]$	normalizer orders	indices over H	conjugate in $W(E_6)$	exact predicate	note
12/15	4	12960	(192,96)	(48,24)	no	no	
17/21	4	12960	(64,32)	(16,8)	no	no	
29/36	6	8640	(72,48)	(12,8)	no	no	
31/39	6	8640	(72,24)	(12,4)	no	no	
41/42	8	6480	(384,384)	(48,48)	no	no	
46/48	8	6480	(96,96)	(12,12)	no	no	
57/58	8	6480	(64,32)	(8,4)	no	no	
59/64	8	6480	(32,32)	(4,4)	no	no	
112/120	16	3240	(64,64)	(4,4)	yes	no	near miss
132/140	18	2880	(216,72)	(12,4)	no	no	
301/303	162	320	(324,324)	(2,2)	yes	yes	qualifier

The exact predicate is `normalizers_conjugate_in_W AND normalizer_indices_over_subgroups=[2,2]`. It is evaluated on every frozen collision bucket; the resulting uniqueness statement has no broader scope.

TABLE 8. Full modular orbit-product fingerprints. “Values” preserves the canonical coset order; “sorted” is an order-independent cross-check.

carrier	values	coefficients	values SHA-256	sorted-values SHA-256	coefficient SHA-256
μ	160	161	4c001e33d82fb9171f1a30 b1e097dba399fa626999bd 00d26ef49912c4cdf50	5ba31ddea225830721d22e 77953f631803e4d5c86368 33e08c05ee1378296cf9	b8818888c1ceb83e05d2f2 df045e9d6e418f1ea18a5f 019d1398e4cd0a59ef6b
ξ_0	320	321	06e82711bbbd9a064e6ab0 8e76b931c2cef90d316919 ff7610fc79efbd98a32d	3ba7bfa6e492d7c83cf244 6a1ab55deb064d6bcca372 dcf5e5e868e34bb122f1	ffe9439cd390729bbb0dd7 ffa4c6a1045c7fbc9c645e 0f37e75c71d1e786e10d
λ	640	641	7812932d7d9119000c19 51e340ac7b23546609f03c 38b8f54158fcf3fc58af	36be0aedaf7b9641a56fed eb4d7b168f0daac1928b1f b963ecdf09db72171674	c82feda40496156b7d006d e4e47a1b808b3cf3ffffe4 a386652d3e3fa77861f1

TABLE 9. Valuations of the signed field discriminants. All signs are positive.

field	3	5	181	283	997	1801	2346241	q
M	308	248	96	80	96	80	96	80
F_+	624	496	192	160	192	160	192	160
F_0	632	496	192	160	192	160	192	160
F_3	624	496	192	160	192	160	192	160
L	1264	992	384	320	384	320	384	320

TABLE 10. Absolute 3-adic rows for $D_3 = \text{ToM } 140$.

field	rows $(n, e, f, d)^{\text{mult}}$	factors	degree	different
M	$(1, 1, 1, 0)^4, (3, 3, 1, 5)^6, (6, 6, 1, 11)^2, (9, 9, 1, 18)^6, (18, 18, 1, 37)^4$	22	160	308
F_+	$(1, 1, 1, 0)^8, (6, 6, 1, 11)^{10}, (9, 9, 1, 18)^8, (18, 18, 1, 37)^{10}$	36	320	624
F_0	$(2, 2, 1, 1)^4, (6, 6, 1, 11)^{10}, (18, 18, 1, 37)^{14}$	28	320	632
F_3	$(2, 2, 1, 1)^4, (3, 3, 1, 5)^{12}, (6, 6, 1, 11)^4, (9, 9, 1, 18)^4, (18, 18, 1, 37)^{12}$	36	320	624
L	$(2, 2, 1, 1)^8, (6, 6, 1, 11)^{20}, (18, 18, 1, 37)^{28}$	56	640	1264

TABLE 11. Absolute 3-adic rows for $D_3 = \text{ToM } 206$.

field	rows $(n, e, f, d)^{\text{mult}}$	factors	degree	different
M	$(2, 1, 2, 0)^2, (6, 3, 2, 5)^3, (12, 6, 2, 11)^1, (18, 9, 2, 18)^3, (36, 18, 2, 37)^2$	11	160	308
F_+	$(2, 1, 2, 0)^4, (12, 6, 2, 11)^5, (18, 9, 2, 18)^4, (36, 18, 2, 37)^5$	18	320	624
F_0	$(4, 2, 2, 1)^2, (12, 6, 2, 11)^5, (36, 18, 2, 37)^7$	14	320	632
F_3	$(4, 2, 2, 1)^2, (6, 3, 2, 5)^6, (12, 6, 2, 11)^2, (18, 9, 2, 18)^2, (36, 18, 2, 37)^6$	18	320	624
L	$(4, 2, 2, 1)^4, (12, 6, 2, 11)^{10}, (36, 18, 2, 37)^{14}$	28	640	1264

TABLE 12. Complete collected relative local table for the first retained decomposition-group branch. A base row is (n, e, f, d) ; each field column is (g, e, f, d) relative to that base prime. Multiplicity expansion recovers every base prime, so no local row is suppressed.

base (n, e, f, d)	F_+ (g, e, f, d)	F_0 (g, e, f, d)	F_3 (g, e, f, d)	L (g, e, f, d)	multiplicity
$(1, 1, 1, 0)$	$(2, 1, 1, 0)$	$(1, 2, 1, 1)$	$(1, 2, 1, 1)$	$(2, 2, 1, 1)$	4
$(3, 3, 1, 5)$	$(1, 2, 1, 1)$	$(1, 2, 1, 1)$	$(2, 1, 1, 0)$	$(2, 2, 1, 1)$	6
$(6, 6, 1, 11)$	$(2, 1, 1, 0)$	$(2, 1, 1, 0)$	$(2, 1, 1, 0)$	$(4, 1, 1, 0)$	2
$(9, 9, 1, 18)$	$(1, 2, 1, 1)$	$(1, 2, 1, 1)$	$(2, 1, 1, 0)$	$(2, 2, 1, 1)$	2
$(9, 9, 1, 18)$	$(2, 1, 1, 0)$	$(1, 2, 1, 1)$	$(1, 2, 1, 1)$	$(2, 2, 1, 1)$	4
$(18, 18, 1, 37)$	$(2, 1, 1, 0)$	$(2, 1, 1, 0)$	$(2, 1, 1, 0)$	$(4, 1, 1, 0)$	4

This collected table is complete: expanding each row by its multiplicity gives all base primes in the first retained local branch. Here g is the number of relative factors over the displayed base prime.

TABLE 13. Complete collected relative local table for the second retained decomposition-group branch, using the same tuple convention as the first branch. Multiplicity expansion recovers every base prime; the table does not select this branch.

base (n, e, f, d)	F_+ (g, e, f, d)	F_0 (g, e, f, d)	F_3 (g, e, f, d)	L (g, e, f, d)	multiplicity
$(2, 1, 2, 0)$	$(2, 1, 1, 0)$	$(1, 2, 1, 1)$	$(1, 2, 1, 1)$	$(2, 2, 1, 1)$	2
$(6, 3, 2, 5)$	$(1, 2, 1, 1)$	$(1, 2, 1, 1)$	$(2, 1, 1, 0)$	$(2, 2, 1, 1)$	3
$(12, 6, 2, 11)$	$(2, 1, 1, 0)$	$(2, 1, 1, 0)$	$(2, 1, 1, 0)$	$(4, 1, 1, 0)$	1
$(18, 9, 2, 18)$	$(1, 2, 1, 1)$	$(1, 2, 1, 1)$	$(2, 1, 1, 0)$	$(2, 2, 1, 1)$	1
$(18, 9, 2, 18)$	$(2, 1, 1, 0)$	$(1, 2, 1, 1)$	$(1, 2, 1, 1)$	$(2, 2, 1, 1)$	2
$(36, 18, 2, 37)$	$(2, 1, 1, 0)$	$(2, 1, 1, 0)$	$(2, 1, 1, 0)$	$(4, 1, 1, 0)$	2

This collected table is complete: expanding each row by its multiplicity gives all base primes in the second retained local branch. Here g is the number of relative factors over the displayed base prime.

TABLE 14. Evidence bindings for the paper-facing figures and tables. Each digest is recomputed from an authoritative artifact or from the exact formal-root ledger and checked against the live Route and independent report. These bindings certify source identity without implying paper or release completion.

field	source-owned value
role	integrated certificate
artifact	c60_certificate.json
SHA-256	d325de1bb0388ccc0c2e81d41fbc6c8fffd692ff777f23647d9e88367d6c2518
binding	Route + independent check
role	independent group replay
artifact	c60_group_evidence.json
SHA-256	dcdb9a8be954d4ea5376220d55fcb9ae9bbb08eb49d03d98d57d790c319ad5fb2
binding	PASS; rebound by checker
role	independent resolver replay
artifact	c60_resolver_evidence.json
SHA-256	f115125725c9160ee3d02f1996147098c234226bdc81eaa670460802a8d827da
binding	PASS; rebound by checker
role	independent check report
artifact	c60_check_report.json
SHA-256	25bc9c1c656da742359814054b66c05e18a304ca85741776c055152a30a98e44
binding	PREFREEZE_CODE_RESULTS_PASS
role	strict certificate schema
artifact	c60_schema.json
SHA-256	c7ddb4ff8fa890f9f801d615158c9038299487affa3808f25fe5d73c987791a5
binding	parsed deep-equal
role	scoped machine manifest
artifact	scoped_hash_manifest.json
SHA-256	f8d44a1929b6f873d4f1b4e7317222c0f06e927ba1977f00f493b8fb004cfec7
binding	self-excluding; Route-bound
role	post-machine formal package
artifact	13-root SHA-256 ledger
SHA-256	44b58dc4b43732803dc7b4b04bd0f86673d2161bf95afc9f8f74a77e048bd7a2
binding	FORMAL_DOCS_PASS

The formal-package digest is the SHA-256 of the literal basename-sorted ledger of the exact post-machine root documents. The Route is excluded from that aggregate to avoid a self-hash cycle. This evidence ledger records reproducibility authority; it is not a release manifest.

TABLE 15. Paper claims and their independent finite checks. The named gate supplies a premise; the proof location supplies the mathematical implication.

claim	gate	principal artifact	independent check
C1 lattice/fields	G1,G4	group evidence	exact arrays, cores, normalizers, meet/join, quotient
C2 primitive fields	G2	resolver evidence	stabilizers plus 160/320/640
C3 invariant gap	G3	both component evidence files	split values
C4 uniqueness	G1	group evidence	full point/pair partitions and cubic stabilizer
C5 character/zeta	G4	group evidence	all eleven rows, no target filter
C6 global arithmetic	G5	group evidence and released input	all 25 class values
C7 local towers	G6	group evidence	all orbit counts, support, tower differences
C8 reproducibility/scope	G0,G7	certificate, schema, check report, manifest	every absolute and relative row in both branches
			independent full-leaf rebuild, mutation and snapshot suites